## FiveAtlas — Instructions

1. On Windows: extract the zip file, then in the extracted folder double click FiveAtlas.exe and it will start running! The first time you run it, you might get a Windows error saying that it does not trust this app. Just press “More info” and then “Run anyway.” Copy the folder to your own computer first — do not run it straight from the shared drive.

2. On a Mac: it needs an Apple Silicon Mac (M1, M2, M3, M4…) on macOS 12.3 or newer — click the Apple menu, About This Mac, and check it says “Chip: Apple M…”; if it says Intel it will not open. Download FiveAtlas-…-macos-arm64.dmg on the Mac itself (passing it through a Windows computer or the shared drive can break it). Double click the .dmg, drag FiveAtlas onto the Applications folder in that window, then eject the disk image. Always start it from Applications.

3. The first time you open it on a Mac, macOS will refuse: “FiveAtlas is damaged and can’t be opened” or “Apple could not verify FiveAtlas is free of malware”. That is Apple’s Gatekeeper reacting to an app that did not come from the App Store, not a problem with the app, and it only happens once per download. The fix that always works: press Cmd+Space, type Terminal and press Return, then paste this one line exactly and press Return: xattr -dr com.apple.quarantine /Applications/FiveAtlas.app Nothing is printed when it worked. Now open FiveAtlas again. (On macOS 15 you can instead open System Settings, Privacy & Security, scroll down to where it says FiveAtlas was blocked, and press “Open Anyway”.)

4. FiveAtlas has no window of its own. On a Mac, a few seconds after you open it (up to a minute the very first time, while macOS checks it) a new tab opens in your browser at http://127.0.0.1:8050 — if no tab appears, type that address into your browser yourself. It keeps running while you work, even if you close the tab for a while. To stop it, just close the tab — it stops by itself about ten minutes later. On Windows the black window that opens is the app: keep it open while you work and close it to stop.

5. Updating: from 1.3.3 on, FiveAtlas updates itself. When a newer version is out it downloads in the background, and a bar at the top of the window offers Restart to update — one click, and the new version opens in a new tab. Not clicked? It switches the next time you open FiveAtlas. Nothing is replaced while you work, and you are asked first if you have unsaved edits. If an update can’t be installed, the line at the top of the sidebar links to the download page instead, which has the Windows and the Mac download on one page. The version you are running is shown in the top-left corner of the window. To update by hand (or from a version older than 1.3.3): on Windows, unzip the new version and open its FiveAtlas.exe — it starts in its own tab even if the old one is still open; close the old black window when you are done with it. On a Mac, drag the new FiveAtlas into Applications, choose Replace, and run the Terminal line above once more. If the old one is still running, its tab says the new version has been installed and offers Restart FiveAtlas; with no tab open, it steps aside within a minute, so opening the app starts the new one. (Updating a Mac from 1.3.0 or older: quit the old FiveAtlas in Activity Monitor first, or restart the Mac — those versions cannot be switched automatically.)

6. Your data lives on the shared drive. On a Mac, connect to it in the Finder first (Go, Connect to Server). Then press Open folder at the top of FiveAtlas and click on the folder that contains the morphology_focus folder, morphology.ome.tif, transcripts.zarr.zip, etc. All of these files should

already be in one folder based on how Xenium exports their data. Before anything is loaded, a screen lists every file the app can load from that folder, by its real name: the imagery, the stains, transcripts.zarr, each region file on its own line, the annotation notes and the metadata file. Everything is ticked — untick anything you don't want and it is not read at all, then press Load. Tick one region file or several (they load together). If you prefer to paste a path on a Mac, it must start with /Volumes/… (not smb://…).

7. To bring in a different region file later, open Files & export in the sidebar and press "Load regions file" — it takes GeoJSON, or a CSV of vertices (one row per vertex with name,x,y columns; Xenium's micron column names are converted automatically). The file dialog may take a few seconds; if it does not pop up, look at your task bar (Windows) or your browser (Mac). The file is read first and nothing changes until you choose: Add to the map keeps what is already there and puts the new regions beside it (a name that is already used gets changed, so two regions never share one), and Replace the map swaps everything for the file. If the damage tools are on (Settings) and the file contains hemi, you are asked first whether to load the damage with it — Yes or No — which covers damage shapes inside that file and damage-only files (like separation.4.geojson) in the same folder. Ctrl+Z undoes the load, and nothing is written until you Save.

8. Predict regions (top bar, beside the dataset) draws the brain regions of the open section from its transcripts, with the region model that comes with the app. Under ■ say which side of the section is medial and which way is up as it lies on screen, and pick an outline (usually hemi) to predict inside. It takes a few seconds to a minute; the transcripts start loading as soon as the dataset opens, so it is quicker if you open the dataset first and look around. When it is done you choose Replace (the regions on screen go, your hemi outline and damage stay) or Add alongside. Ctrl+Z undoes it. The regions are named the way annotators name them (ISO, HPF, TH, CP, HY, PAL, OLF, CTXsp, sAMY, VL, V3, opt); ft is every other fiber tract, dft and lft together, for you to cut in two with Split. It is a starting point to check and edit, not a finished map.

9. If a region file uses one name for more than one shape, FiveAtlas asks about it as soon as the dataset opens (and after Replace the map or Reset): it lists each such name, with whether the shapes are identical or how much they differ. Rename them keeps the name on the first shape and makes each other one its own region, numbered .1, .2 … (PAL becomes PAL.1). Make each one region makes each name one region: identical shapes become one, and shapes that differ are combined into one outline covering the ground either of them covered. Not now leaves everything as it is. Ctrl+Z undoes either answer, and nothing is written until you Save.

10. If you are ever unsure about how to use certain features, there are instructions that pop up after you click on a feature. The sidebar is in sections — What loaded, Regions, Edit regions, Files & export and more — and clicking anywhere on a section's title bar folds that section away so the sidebar stays short; click again to bring it back. Layers, Stains and Genes have their own sidebar on the right, so they are always in reach; Hide » tucks it away and the tab at the right edge brings it back. Either sidebar can be made wider or narrower by dragging the thin strip between it and the image. Settings (the gear button at the top of the sidebar) makes the text and buttons bigger or smaller.

11. The Stains panel (in the right-hand sidebar) lists every morphology_focus channel (DAPI, 18S, alphaSMA/Vimentin…). Tick a channel to show it, pick its colour, and set its brightness with the min and max sliders — min is the level that reads as black, max the level that reads as full colour, so

narrowing the range brightens the faint staining in between. The γ (gamma) slider bends the curve in between: below 1 lifts the faint staining further, above 1 darkens it. The image updates while you drag. Gene channels have the same controls.

12. Gene expression has two looks, switched at the top of the Genes panel. Glow is the familiar additive one. Heat map draws the transcripts as perfect squares: pick the square size (10, 20, 40 or 80 microns), and each gene reads light where it is sparse and dark where it is dense — a blue gene runs pale-blue to deep blue, and every square holding any transcripts at all stays visibly tinted. Add a second gene in red and the two mix like inks: where both are dense the squares go dark purple. Or pick one of the scientific colour scales (Viridis, Inferno, Magma, Plasma, Turbo — the same ones Xenium Explorer uses) to paint the combined density of every shown gene on one scale. The min/max sliders still set what counts as sparse and dense.

13. All of your regions are listed in the Regions section of the sidebar. Click one to select it. Ctrl-click (Cmd-click on a Mac) adds another one, and Shift-click takes a whole run — in the list or on the image. With several selected, Delete selected removes them all in one step, and one Ctrl+Z brings them all back. If a name appears twice with “part 1/2” beside it, that is one region made of two separate pieces, and anything you do applies to both — except Delete this piece, below.

14. Right click a region (two-finger click on a Mac trackpad), either in the list or on the image, to rename it, change its colour, bring it to the front, send it to the back, or delete it. Bring to front draws it over everything else, so it is the one your clicks land on; Send to back draws it behind everything else, so whatever it contains stays visible. On a region made of separate pieces, right-clicking one piece on the image also offers Delete this piece: only that piece goes, the rest of the region stays, and Ctrl+Z brings it back. Use it for a stray island a region picked up, like a triangle of Caudoputamen off on its own. If the stray piece lies under another region, right-click it anyway: the menu is the top region's, and it also offers Delete this piece of (the region underneath).

15. The use button beside each region in the list switches it off: it is hidden from the image and left out of gap-finding and Check geometry, and a new drawing does not cut into it, but it stays in the file and comes back exactly as it was. You no longer need to switch off the outline (“hemi”) to hunt for gaps: the app sets it aside by itself for Fix a gap, Share borders and Check geometry.

16. Beside it, every region also has a face toggle and a border toggle. These only change what you SEE — unlike use, the region still counts for gap-finding, snapping and Check geometry. Hide a face to see the imagery and the borders clean while you fill gaps; hide a border to see a fill without its outline. The Faces and Borders buttons above the list do every region at once (each says what a click will do), and Show all turns every face, border and switched-off region back on. Hold H to peek at the bare image with every region hidden, and press Home to fit the whole section. Type in the filter box to find a region; double-click its row to zoom to it, or right-click it for Show only this. Under Display, two sliders set how solid the region faces are and how thick the outlines are drawn.

17. Edit points reshapes a region by hand. Select the region, press Edit points, then drag the points along its outline. Click the outline between two points to add one. Right click a point to delete it: the rest of that border re-spaces itself so no long empty stretch is left, and a point on a border shared with a neighbour is removed from both regions, so the border never opens up. A point where three regions meet, or where a shared border ends, is protected and the app says why. Tick Proportional

editing (at the bottom of Edit regions) and the nearby points will follow the one you are dragging, and you can scroll your mouse wheel while dragging to change how much of the outline comes with it. On a touchpad, set that size before you grab a point: drag the Radius slider, press [ and ], or hold Alt and scroll over the image — the ring shows how far it reaches.

18. When you open Edit points on a region, its points are tidied automatically: you get fewer points with none bunched together, the outline barely moves, and every neighbour's side of a shared border is re-pointed with it, so no border opens up and no gap appears. Points that still sit right next to another are then deleted the way right-click Delete point deletes them — from both regions on a shared border — except where regions meet. A note over the map says what changed ("Tidied: TH 127 → 51 points"); press Undo (or Ctrl+Z) to put the old points back. The same tidy happens by itself right after Share borders, Merge, Fix a gap, Clean up stray lines, Build the outline and drawing a new region, as its own Undo step. Points that still sit right next to another are drawn orange. Tidy in the Tidy points box runs it by hand. Untick Tidy automatically (under Adjust…) to only ever tidy by hand. Adjust… lets you choose the number yourself: drag the slider from more to fewer or type a number and press Enter (a number it can't reach goes to the closest one), and tick Space points evenly to put the points the same distance apart all the way round. The slider goes all the way down to 3 points. When a count would change the shape it says "The shape will deform" — check the outline, then press Keep to keep it or Cancel to put the old points back. Below the fewest points that keep every border joined, only that region changes, so its borders no longer meet its neighbours' exactly; Share borders joins them again.

19. Move region drags a whole region to a new place without changing its shape. Select the region, press Move region, and drag it on the image. Nothing is trimmed, so it may end up overlapping a neighbour — use Bring to front or Send to back from the right-click menu, or Share borders to tidy the join.

20. Edit shared borders works on the line between two regions. Click two regions that touch and you can drag the border between them, and both sides update together so you never open up a gap. Right click the red border line to add a point there, or right click a point on it to delete it. This works for a region INSIDE another region too: pick the pair and the inner one's outline appears as their border — drag it, and the outer region keeps wrapping it — and pressing Share borders on such a pair merges the inner region's edge into the outline where they neighbour, so they genuinely share it. To pick a region that sits underneath everything (like "hemi"), click its NAME in the Regions list. A drag may also reach PAST the section outline — the region grows into the empty ground beyond it (rebuild "hemi" afterwards if you want the outline to catch up). Other regions' territory is never taken this way.

21. Pick two or more regions and choose where the shared border goes: Middle moves both outlines to meet halfway, and Snap to (a region) leaves that region exactly as it is and brings the other onto its edge. Either way all the boundaries are tidied at once: small gaps get filled, overlaps get removed, and every pair ends up with one shared edge. Fill all pixels in the gap (ticked to start with) fills every free pixel; untick it to leave empty the strips beside a region standing in the gap. The Snap reach slider sets how wide a gap Share borders will bridge — while you pick, it tells you how far it has to reach for the two regions you picked. If a gap is left, raise it and press Share again; a second press is always safe. Click a picked region again to unpick it. Merge into one instead combines everything you picked into a single region. If "hemi" is among the picks, Merge asks first: Fold into hemi

removes the other regions and gives their ground to the outline, or Cancel.

22. Merge into one truly dissolves the border between the regions you picked: the shared line disappears and they become a single region — even when a hairline gap ran along the old border, it is sealed. Pieces that genuinely do not touch stay as separate parts of the one region; the app never invents tissue between them.

23. Once you have shared borders, the border's points are tidied automatically the same way (with Undo), and Tidy points appears there too: press Tidy, or Adjust… to choose the number, then Keep or Cancel. This is useful when a border has far more points on it than you can work with.

24. Split a region cuts one region into two. Press Split a region, then draw a line all the way across the region on the image, clicking for each point, and right click (or press Enter) to finish. You don't need to pick the region first: the line finds it, and if it could split more than one region, the app asks which. Press Esc to throw the line away and start again. To separate a piece that is already apart from the rest, draw the line through the gap between them instead: the pieces on each side become their own region, and the larger side keeps the name.

25. Add a new region or damage lets you draw a region from scratch. Choose “Anatomical region” under “What are you drawing?”, then trace the outline on the image, clicking for each point, then right click or Enter to finish and Esc to cancel. If your outline lands on another region, you choose what happens. Replace: the new region takes that ground from the regions already there. Overlap: both keep their whole shape — a region drawn around another goes behind it, so the smaller one stays visible. Fit around: the new region is trimmed to the free ground, and nothing already there changes. Cancel throws the outline away. (A choice that would erase a region, or leave nothing of your shape, is not offered.) A region you have switched off is left whole: your outline overlaps it instead of cutting into it.

26. The damage tools (drawing damage, the damage cells and Annotation notes, below) are hidden unless you tick Show damage & notes tools in Settings; your files keep their damage either way. With them on, the same button draws damage. Pick a damage type instead (Separation, Large void, Bubble…) and trace the damaged area the same way. The shape names itself — separation.1, voidlarge.2 — with numbering running across the whole file. Unlike a region, a damage shape sits inside the region it covers: nothing is taken away, and the two overlap on purpose. If a shape reaches into more than one region, you are asked which region it belongs to — the one holding most of it is suggested — or record it in both if two annotators each own one side. Some damage types (Cutoff, Missing, Low/no transcripts) are never drawn; you tick those per region in the damage cells panel instead (see below).

27. Build the hemisection outline creates (or rebuilds) “hemi”, the region that wraps around all the others, computed from the regions themselves rather than traced by hand. You see the outline and its area as a preview first, then press Create hemi or Replace hemi. Damage shapes are left out, rebuilding never uses the old outline as input (so it cannot creep outwards), and a rebuilt hemi keeps its colour and anything you had ticked on it.

28. Fix a gap is for empty spaces left behind between regions. Click inside one and it will be outlined in red, then press Dissolve to hand it to the regions around it, or New region to turn it into a region of its own. Click another gap to carry on; Esc leaves Fix a gap. A hole inside a single region, and the

margin between the tissue and hemi's outline, are not gaps and are refused.

29. Clean up stray lines removes the hairline slivers geometry operations sometimes leave behind — too thin to click, too thin for Fix a gap. Hold the left button and sweep a circle around them, then release. Everything sliver-thin inside the circle is found and highlighted in red with a report of where it came from and where the ground will go; press Apply to remove it, Cancel to keep everything, Esc mid-sweep to throw the circle away. Healthy regions are never touched — even if your circle crosses straight through one — and neither are damage shapes.

30. Orientation is for a slide that was imaged upside down or mirrored. Rotate or flip the whole thing — image and regions move together — and work the way round that makes sense. When you export, you choose whether the file keeps the displayed orientation (After rotation, offered first) or is turned back to the as-imaged one.

31. Undo and redo are the two curved-arrow buttons at the top right of the window, or Ctrl+Z and Ctrl+Y (Cmd+Z and Cmd+Shift+Z on a Mac), and they work for everything above.

32. Save is at the top right too, next to undo and redo. "● unsaved" beside it means you have edits that are not on disk yet, and "saved" means everything is. The first time you press Save on a dataset, it asks where to put the regions file — it opens in the dataset's own folder with a name already filled in, like merged.regions_edited.geojson. Ctrl+S saves too. Every Save after that updates the same file without asking: "Saving to (file)" shows which one, and (change) picks a different file. After each Save a panel under the button shows exactly where it went, with Open folder and Copy path buttons. It never overwrites the dataset's original region files — if you pick one, it offers a new name beside it instead.

33. Every save also keeps a dated copy in the app's own folder (%LOCALAPPDATA%\FiveAtlas on Windows; ~/Library/Application Support/FiveAtlas on a Mac — in Finder press Shift+Cmd+G and paste that path), so you can always go back to an earlier one; the log file lives there too. Reset, beside Save, asks first and then throws away every edit since the last save (Ctrl+Z still brings them back). Closing the window or opening another dataset with unsaved edits asks first too. Restore original GeoJSON (in Files & export) takes you back to the dataset's own file — your current work is snapshotted first, so even that can be undone. Your original files are never changed.

34. Check geometry will tell you if anything is wrong with your outlines, and offers to repair the ones it can fix on its own. Regions you have switched off are left out of the check.

35. Damage cells, region by region is for the SmartSheet. It shows one box per region with that region's damage, using the sheet's own dropdown names. Tick Done, or add a damage type by hand — this is where the never-drawn kinds (Cutoff, Missing, Low/no transcripts) are recorded. Copy a box, click that region's Damage cell in SmartSheet, and paste: it stays in one cell. A ◆ marks damage that came from a shape you drew — delete the shape to remove it.

36. Annotation notes writes what you found into the sample's shared files (annotation.notes.yaml and metadata.yml, in the dataset folder). Type your name in Annotator — left blank, no name is written and only damage and voids are. Choose Only what I edited or All regions, then press Preview notes changes: every line that would change is shown as a diff before anything is written, and nothing is written until you press Save in that panel. These files are shared, so the app protects them: if a

colleague saved after your preview, your save is refused rather than overwriting them, and damage already recorded in the file is never cleared just because your working copy has no shape for it. If the sample has no notes file yet, Create it here makes one — but check no one has started one in another folder for the same sample first.

37. When you are done, press Export. Merged gives you one .geojson with every region in it, and Separate gives you a zip with one file per region. AnnData (.h5ad) exports a regions × genes count table for scanpy — the transcript density summed inside each region, with areas and centroids — ready for Python analysis.